ADIDAS US SALES ANALYTICS DASHBOARD

Project Report Submitted by	Pranjal Gupta
Domain	Adidas US Sales Analytics Dashboard
Tools & Technologies Used	Excel | Python |Power BI
Techniques Applied	Data Cleaning, Exploratory Data Analysis (EDA), Data Aggregation, Data Visualization, Data Modeling, Time Series Analysis, Sales Forecasting, DAX Calculations, Interactive Dashboard Design.
Project Difficulty Level	Intermediate

Dataset Availability

The cleaned dataset used for this project is publicly available in the GitHub repository:

Dataset Link: https://github.com/pranjalgupta4482-ui/adidas-sales-analytics-dashboard/blob/main/data/raw_data/Adidas%20sales%20analytics.xlsx

Dataset Information

The dataset used in this project contains **Adidas sales data** that provides information about product sales across different regions,

retailers, and sales methods. The data helps analyse sales performance, revenue trends, and product demand.

The dataset includes transactional sales records with details about products, pricing, units sold, and profitability. It allows analysis at multiple levels such as regional performance, product category performance, and SKU-level sales insights.

Key Columns in the Dataset

- **Retailer** – Name of the retailer selling Adidas products
- **Region** – Geographic region where the sales occurred
- **State** – Specific state of the sale
- **Product** – Name of the Adidas product
- **Product Category** – Category of the product (e.g., footwear, apparel)
- **Sales Method** – Sales channel such as online or in-store
- **Units Sold** – Number of units sold
- **Price per Unit** – Selling price of each unit
- **Revenue (Total Sales)** – Total revenue generated from the sale
- **Operating Profit** – Profit generated from the sale
- **Date** – Date of the transaction

The dataset enables comprehensive analysis of **sales performance, profitability, and market trends**, helping to derive meaningful business insights and support data-driven decision making.

Problem Statement

Retail companies generate large amounts of sales data from multiple regions, products, and sales channels. However, without proper analysis, it becomes difficult to understand which products perform best, which regions generate the highest revenue, and how sales trends change over time.

The lack of clear visualization and structured analysis makes it challenging for businesses to identify key performance drivers and make informed decisions.

Additionally, businesses often struggle to anticipate future sales performance without proper forecasting techniques. The absence of clear dashboards and predictive insights makes it challenging for decision-makers to monitor performance and plan future strategies.

Therefore, there is a need to analyse Adidas sales data, visualize key performance metrics through an interactive dashboard, and incorporate sales forecasting to better understand historical trends and predict future revenue patterns.

Project Objectives

1. To analyse Adidas sales data to understand overall business performance and revenue trends.
2. To identify top-performing product categories and SKUs contributing the most to total sales.
3. To analyse revenue contribution by product category and individual products to determine top-selling items.
4. To evaluate regional sales performance by analysing sales across states and cities to identify high-performing markets.
5. To compare sales performance across different retailers to evaluate retailer-wise revenue contribution.
6. To analyse sales trends over time to understand seasonal patterns and market demand.
7. To develop an interactive Power BI dashboard that visually represents key sales metrics and insights.
8. To implement sales forecasting techniques to predict future revenue trends based on historical sales data.

9. To transform raw sales data into actionable insights that support data-driven decision making.

Data Cleaning and Preprocessing

Before performing analysis and building the Power BI dashboard, the dataset was cleaned and pre-processed to ensure accuracy and consistency.

The following steps were performed:

1. **Handling Missing Values** – Checked the dataset for missing or null values and ensured the dataset was complete for analysis.
2. **Removing Duplicate Records** – Duplicate rows were identified and removed to avoid incorrect calculations and ensure data integrity.
3. **Data Type Formatting** – Converted columns such as date, numerical values, and categorical fields into appropriate data types for proper analysis.
4. **Data Validation** – Verified revenue, units sold, and pricing fields to ensure calculations were accurate and consistent.
5. **Data Structuring** – Organized the dataset into a structured format suitable for analysis and visualization in Power BI.

These preprocessing steps helped improve data quality and ensured reliable insights during the analysis and dashboard development stages.

Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed using Python to understand patterns, relationships, and trends within the Adidas sales dataset before developing the Power BI dashboard. The analysis was carried out using libraries such as **Pandas, NumPy, Matplotlib, Seaborn, and Plotly**.

The following analytical steps were performed in the notebook:

1. **Data Inspection and Structure Analysis**
   The dataset was loaded using Pandas and initial inspection was performed using functions such as head(), shape(), and describe() to understand the structure, size, and statistical distribution of the data.

```
df = pd.read_excel('/content/Adidas sales analytics.xlsx')
df.head()
```

	Retailer	Retailer ID	Invoice Date	Region	State	City	Product	Price per Unit	Units Sold	Total Sales	Operating Profit	Operating Margin	Sales Method
0	Foot Locker	1185732	2020-01-01	Northeast	New York	New York	Men's Street Footwear	50.0	1200	600000.0	300000.0	0.50	In-store
1	Foot Locker	1185732	2020-01-02	Northeast	New York	New York	Men's Athletic Footwear	50.0	1000	500000.0	150000.0	0.30	In-store
2	Foot Locker	1185732	2020-01-03	Northeast	New York	New York	Women's Street Footwear	40.0	1000	400000.0	140000.0	0.35	In-store
3	Foot Locker	1185732	2020-01-04	Northeast	New York	New York	Women's Athletic Footwear	45.0	850	382500.0	133875.0	0.35	In-store
4	Foot Locker	1185732	2020-01-05	Northeast	New York	New York	Men's Apparel	60.0	900	540000.0	162000.0	0.30	In-store

2. **Column Standardization**
   Column names were converted to lowercase and spaces were replaced with underscores to maintain consistency and make data manipulation easier during analysis.

3. **Date Feature Engineering**
   The invoice date column was converted to datetime format. Additional features such as **month, month name, and year** were extracted to support time-based sales analysis.

```
df[['invoice_date','month','month_name']].head()
```

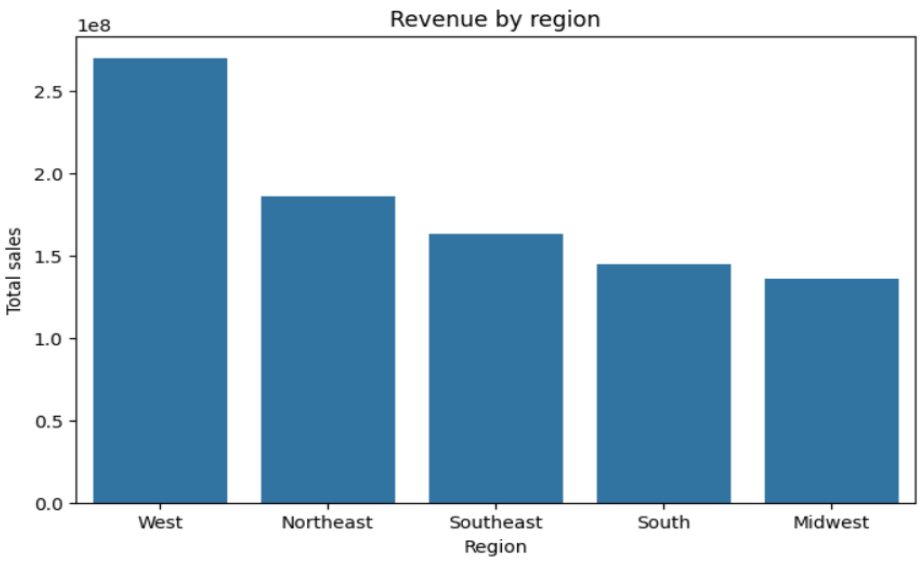
	invoice_date	month	month_name
0	2020-01-01	1	January
1	2020-01-02	1	January
2	2020-01-03	1	January
3	2020-01-04	1	January
4	2020-01-05	1	January

4. **Overall Sales Metrics Calculation**
   Key metrics including **total revenue (total sales), total operating profit, and total units sold** were calculated to understand overall business performance.

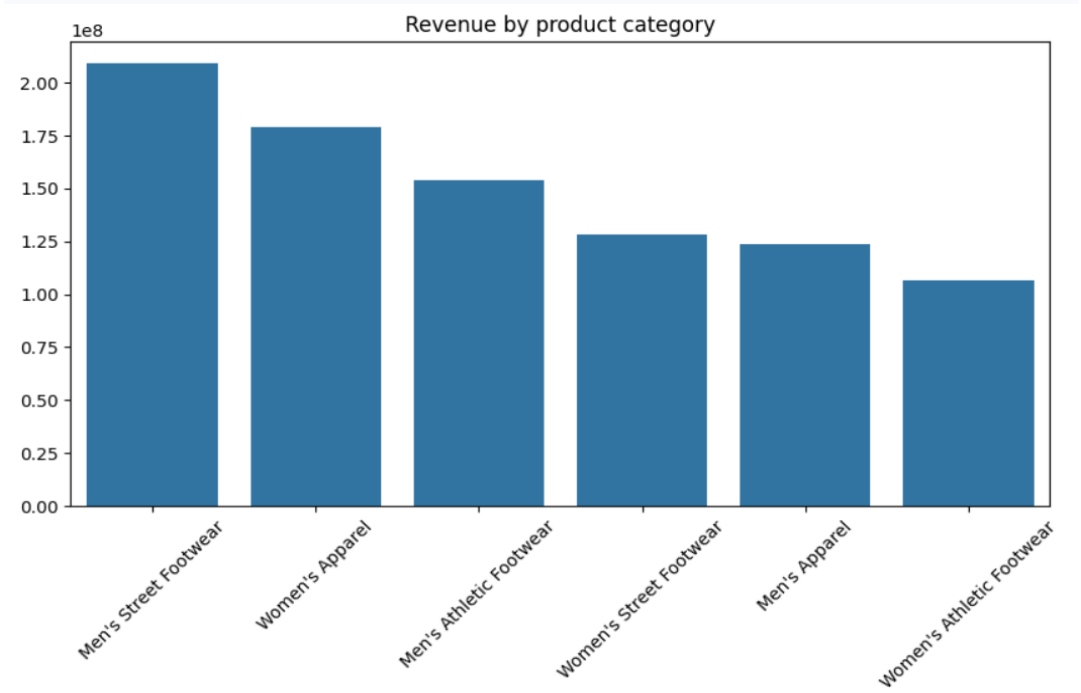
```
total_revenue = df['total_sales'].sum()
total_profit = df['operating_profit'].sum()
total_units = df['units_sold'].sum()

print("Total Revenue:", total_revenue)
print("Total Profit:", total_profit)
print("Total Units Sold:", total_units)
```

```
Total Revenue: 899902125.0
Total Profit: 332134761.45000005
Total Units Sold: 2478861
```

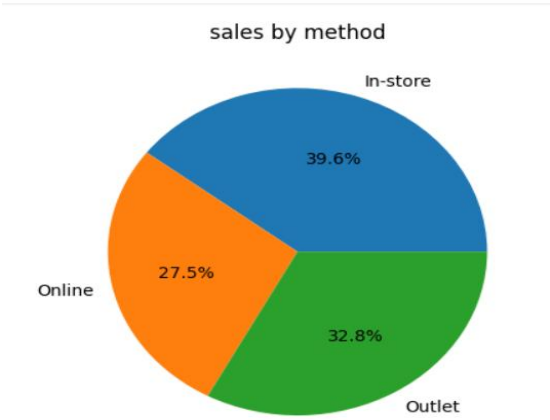
5. **Regional Sales Analysis**
   Sales were grouped by **region** to analyse that west regions generated the highest revenue. Bar charts were created to visualize regional sales performance.



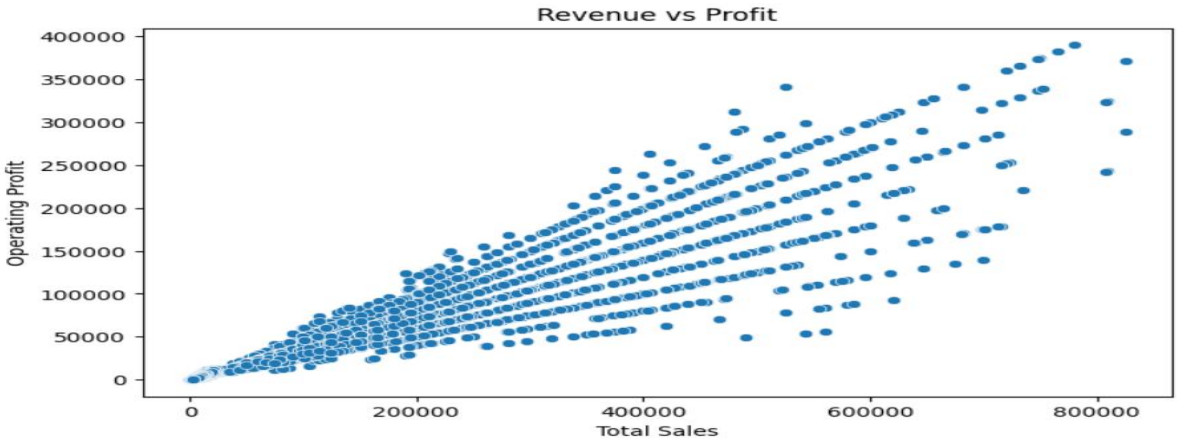
6. **Product Performance Analysis**
   Sales were aggregated by **product** to identify top-performing products contributing the most to total revenue.



7. **Sales Method Analysis**
   Sales were grouped by **sales method** (such as outlet, online or in-store), and a pie chart visualization was created to understand the contribution of each sales channel and thus we get to know that the highest revenue obtained from in-store sales method i.e. 39.6%.

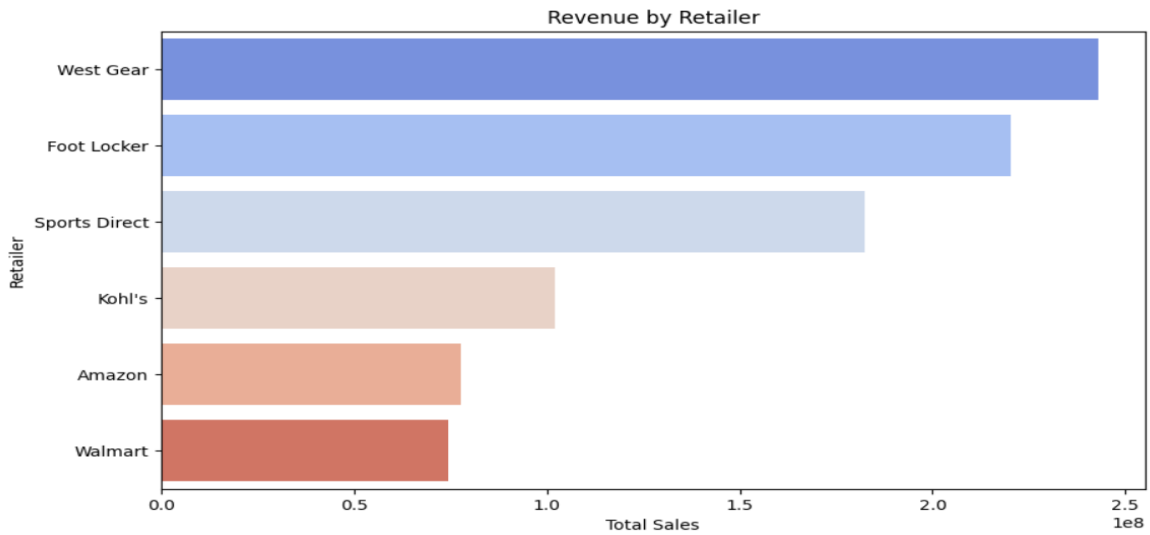


8. **Revenue vs Profit Relationship Analysis**
   A scatter plot was created to visualize the relationship between **total sales and operating profit**, helping to understand profitability patterns.



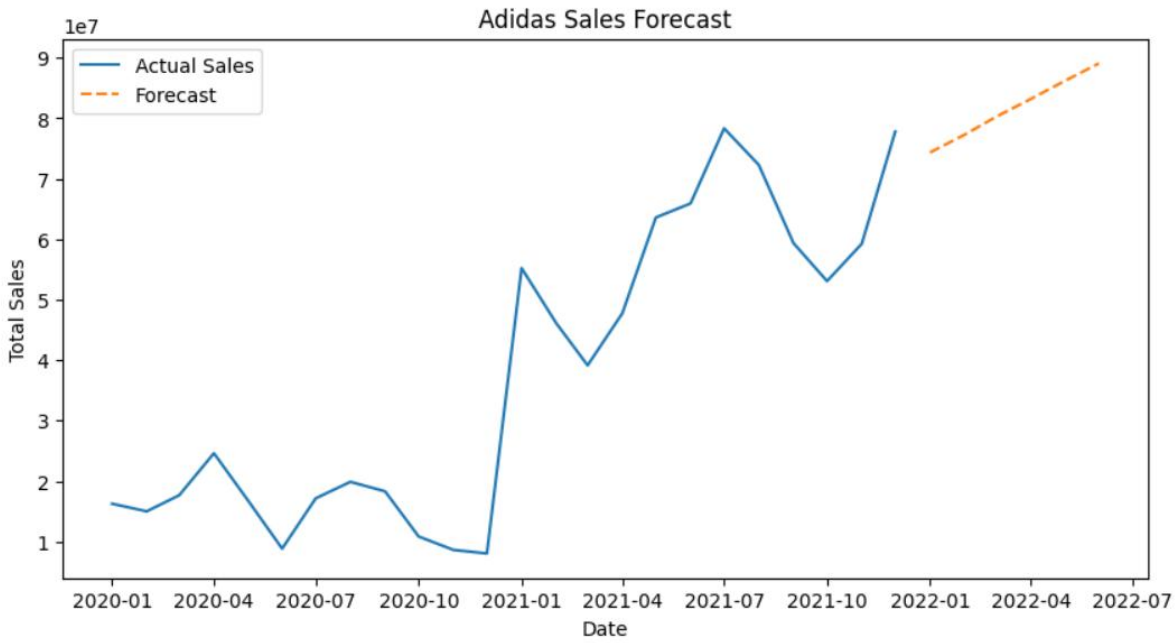
9. **State-Level Sales Analysis**
   Sales were aggregated by **state**, and state abbreviations were mapped to create a **choropleth map using Plotly**, allowing geographic visualization of sales performance across states.

10. **Retailer Performance Analysis**
    Sales were grouped by **retailer** to identify retailers contributing the highest revenue.



11. **Time Series Sales Analysis**
    Monthly revenue was calculated by grouping data by **year and month**, allowing analysis of sales trends over time.

12. **Sales Forecasting Preparation**
    A time index was created and a **Linear Regression model** was used to forecast future sales based on historical monthly revenue data.



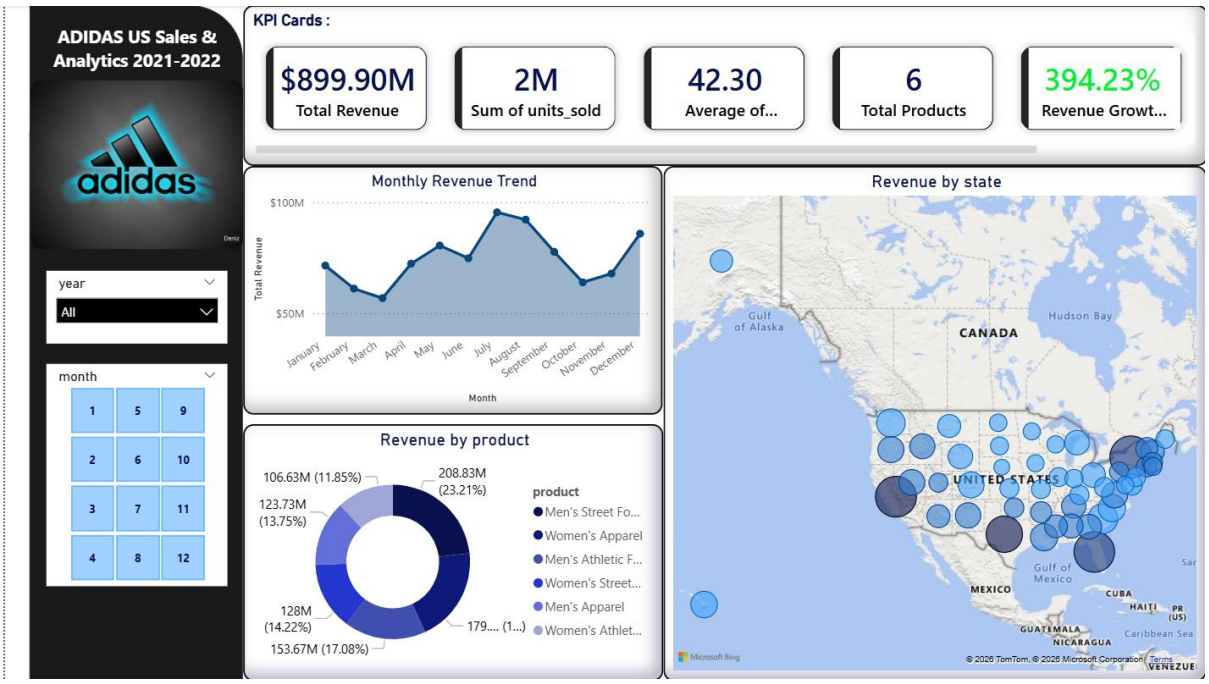
13. **Profit Margin Calculation**
    A new feature called **profit margin** was calculated using operating profit and total sales to measure profitability percentage.

Finally, the cleaned and processed dataset was exported as **"adidas_cleaned_data.xlsx"** for further visualization and dashboard development in Power BI.

Power BI Dashboard Development

After completing data cleaning and exploratory analysis in Python, the processed dataset was imported into Power BI to create an interactive sales analytics dashboard for Adidas US sales performance. The dashboard was designed with multiple pages to provide comprehensive insights into revenue trends, product performance, regional sales distribution, retailer performance, and sales forecasting.

1. Sales Overview Dashboard



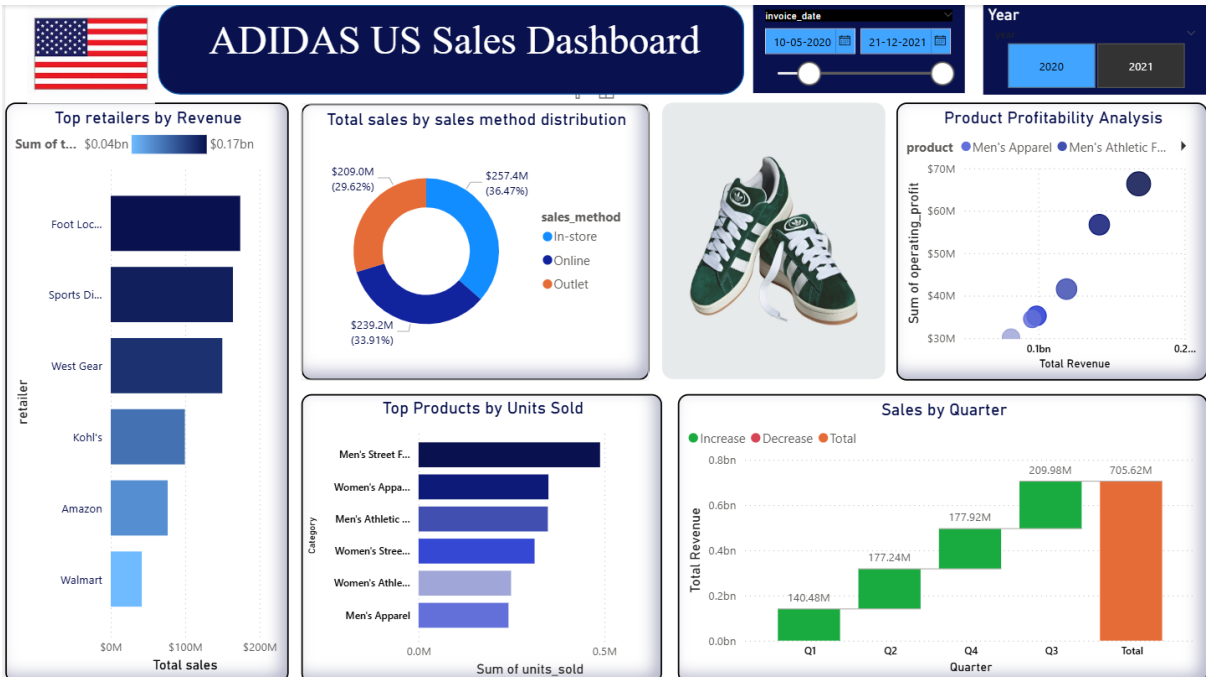
The Sales Overview page provides a high-level summary of Adidas sales performance. Key Performance Indicator (KPI) cards display important metrics such as Total Revenue, Total Units Sold, Average Price per Unit, Total Products, and Revenue Growth Percentage.
A monthly revenue trend line chart was used to analyse sales fluctuations throughout the year and identify seasonal patterns.

A geographical map visualization displays revenue distribution across different US states, allowing users to identify high-performing markets. Additionally, a donut chart showing revenue by product

category helps highlight the contribution of different product segments.

Interactive slicers for year and month allow users to dynamically filter the dashboard and explore data across different time periods.

2. Product and Retailer Analysis Dashboard



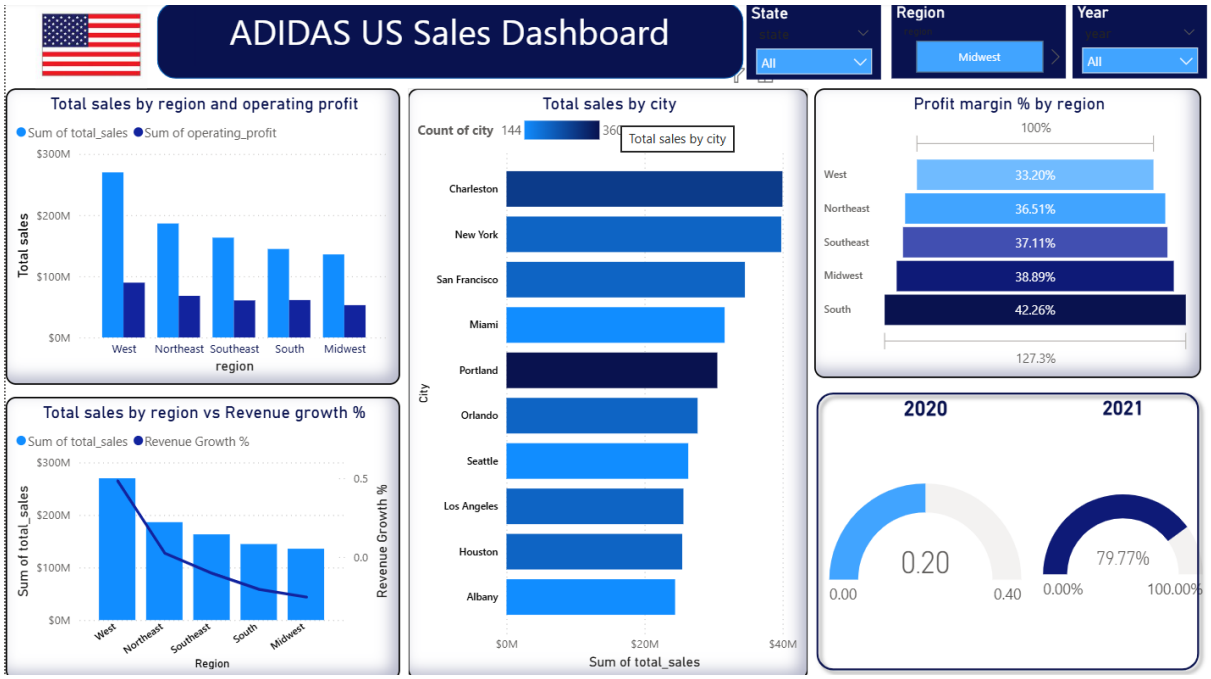
This page focuses on analysing product performance and retailer contributions. A bar chart showing top retailers by revenue helps identify major retail partners driving Adidas sales.

A sales method distribution chart illustrates the proportion of sales generated through In-store, Online, and Outlet channels, providing insights into consumer purchasing behaviour.

A bar chart of top products by units sold highlights the most popular product categories among customers. Additionally, a scatter plot showing product profitability visualizes the relationship between total revenue and operating profit across different product categories.

A quarterly sales waterfall chart further breaks down revenue contributions across different quarters of the year.

3. Regional Performance Dashboard



The Regional Performance page provides deeper insights into geographic sales performance. A clustered bar chart comparing total sales and operating profit by region allows analysis of both revenue and profitability across different US regions.

A city-level sales chart highlights the cities generating the highest revenue, helping identify major sales hubs.

A profit margin analysis by region visualizes the profitability percentage across regions, enabling comparison of operational efficiency.

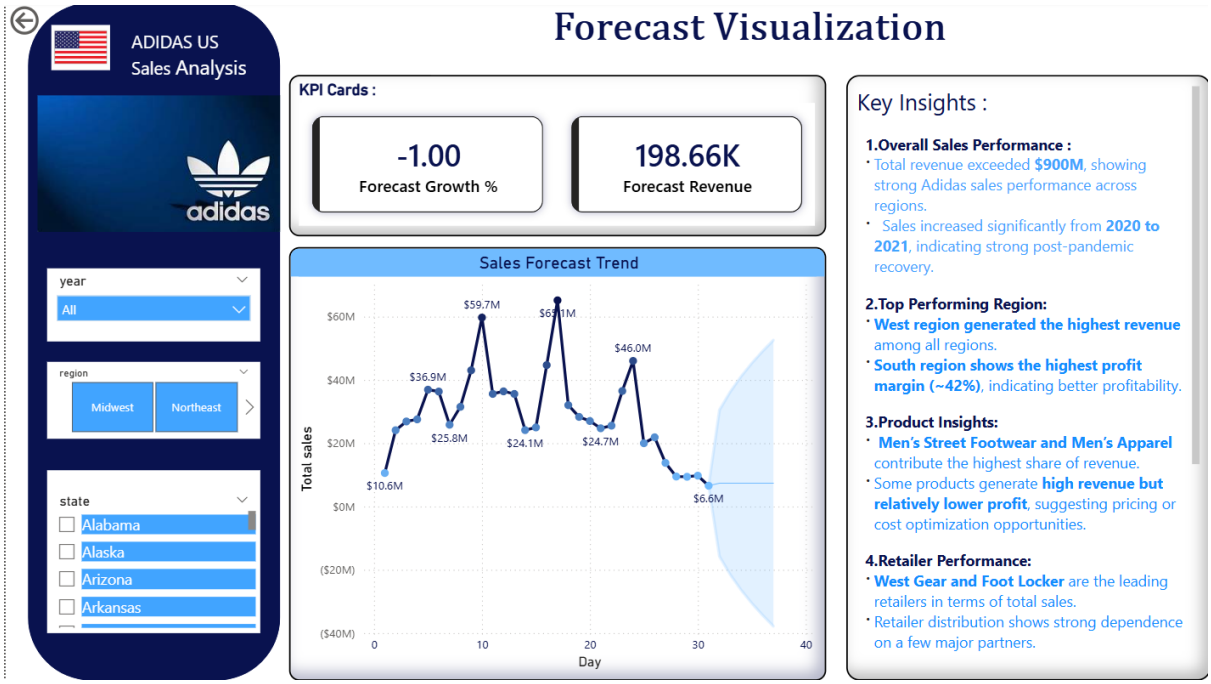
Additionally, a regional sales vs revenue growth chart illustrates how different regions contribute to revenue growth.

4. Forecast Visualization Dashboard

The Forecast Visualization page focuses on predicting future sales trends. KPI cards display forecast revenue and forecast growth percentage, summarizing expected future performance.

A sales forecast trend chart was created using historical sales data to estimate future revenue patterns. The forecast visualization helps

identify potential sales trends and supports strategic planning and demand forecasting.



This page also includes a Key Insights section, summarizing major findings related to overall sales performance, regional strengths, product contributions, and retailer dominance.

Together, these dashboards provide a comprehensive view of Adidas sales performance and enable data-driven decision making through interactive exploration and visual analytics.

Conclusion

The Adidas Sales Analytics project analysed sales data to generate key business insights using Power BI and forecasting techniques. The analysis shows total revenue of approximately **$899.9M** with over **2M units sold**, indicating strong overall performance.

Men's Street Footwear and Men's Apparel were the top-performing product categories. Regionally, the **West** generated the highest revenue, while the **South region** showed the highest profit margin

(~42%). Key cities such as **New York and San Francisco** contributed significantly to total sales.

Foot Locker and West Gear emerged as leading retailers, and **In-store sales (~36%)** contributed the highest share among sales channels.

Forecasting results indicate a slight decline in growth (**-1.00%**), suggesting potential stabilization in future revenue trends.

Overall, the project highlights how data-driven insights can support better business decisions and performance optimization.

Business Recommendations

1. **Focus on High-Performing Products**
   Increase inventory and marketing efforts for top-performing categories such as *Men's Street Footwear and Men's Apparel* to maximize revenue.

2. **Strengthen High-Revenue Regions**
   Invest more in the **West region** to further boost sales, while replicating successful strategies from the **South region** to improve profitability in other regions.

3. **Optimize Retail Partnerships**
   Strengthen relationships with key retailers like **Foot Locker and West Gear**, as they contribute significantly to overall sales.

4. **Enhance Sales Channels**
   Since **In-store sales contribute the highest share (~36%)**, improve in-store experience while also expanding online channels to capture more digital customers.

5. **Prepare for Future Demand Trends**
   Based on forecast insights indicating slight slowdown, businesses should optimize inventory planning and promotional strategies to maintain stable revenue growth.